

The impact of a photo on decisions

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ABSTRACT

Photos are everywhere. They are posted daily on various social networks such as Instagram, Facebook, or Snapchat. Every day, advertising tries to use photos to change behavior. But do photos really change behavior? Photos are non-numerical information. How does non-numerical information affect a decision? I use experiments to show the influence of a photo on the behavior of the participants. These effects are difficult to represent in a traditional Bayesian framework. Galperti called for an economic theory of persuasion that goes beyond Bayesianism. I use this example to introduce the Seesaw model of a decision. With the Seesaw model, such an effect can be implemented in economic theory.

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1. Research of the impact of a photo on decisions

There are several articles in psychology and communication research that measure the influence of a photo on decisions.

Hsee and Rottenstreich (2004, p. 26) found an effect for charitable donations for pandas. Charitable donations were higher when a photo with a panda was used instead of a panda symbolized by dots. Donations did not differ significantly between a photo with one panda and a photo with four pandas. When the pandas were symbolized by 'dots', donations for four pandas were twice as high as for one panda.

Västfjäll et al. (2014) found that the composition of photos of children in Mali affects charitable donations. They found that donations decreased when there was more than one person in a photo. Västfjäll et al. (2014) found donations of 24.9 SEK (Swedish krona) when a photo with one person was shown, 23.4 SEK for a photo with two people and 21.3 SEK for a photo with eight people. Västfjäll et al. (2014) called this effect 'compassion fade'.

Soroka et al. (2016) analyzed the impact of two different photos on the support for military action. One photo showed a soldier with a child ('peacekeeping photo'), the other photo showed a soldier with a weapon ('conflict photo'). The 'peacekeeping photo' led to greater support for the military involvement. Soroka et al. (2016, p. 574) concluded that 'framing implied by photos can have an important impact on support for military engagement'.

Lu and Peng (2024) analyzed 'approximately ten million tweets from Twitter for three social-mediated protests'. They found that 'posts with images and videos generally attracted more audience engagement than their textual counterparts'. The superior effects of visuals were 'generally more pronounced during the ignition phase of the protest than the periods before and after'.

Lagomarsino and Suggs (2018) examined the effects of visual communication on children's healthy food choices. They analyzed children's preferences for apples, broccoli, carrots, fruit, and water. They found that children liked foods 'displayed as cartoons the most' but preferred to consume foods 'represented by photos' (p. 487). They concluded that 'it may be more effective to use photos of healthy foods rather than cartoons or animations' (p. 487).

Septianto et al. (2023) analyzed how photos influence the willingness to buy a novel product ('clean meat'). They found that 'participants reported a greater willingness to try a novel product (...) when the advertisement featured an illustration (vs. a photograph)'. For a 'familiar product' such an effect was not found.

David (1998) found that 'the addition of a representative picture to a news item improved the recall of that item'. Concrete news items benefited more from an addition of a photo than abstract news items.

Jeon et al. (2019) examined the response to obesity and compared a 'headless photo' and a photo of a 'whole being'. They found that the headless photo of obese people 'promoted the dehumanizing perception of and a negative attitude towards individuals with obesity' (Jeon et al., 2019, p. 498). They recommended 'that pictures of headless obese people be discouraged' (Jeon et al., 2019, p. 500).

Wang et al. (2017) researched the impact of 'smile intensity' on photos. They compared the reaction to two photos. One photo showed a 'slight smile', the other photo showed a 'broad smile'. They found that broad smile is perceived 'warmer' but less competent. They compared the reaction of the photos in low-risk and high-risk decisions. They found that 'a broad (...) smile leads to higher purchase intentions in a low-risk context. In a high-risk context, however, a broad (...) smile decreases behavioral intentions via less favorable competence perceptions' (Wang et al., 2017, p. 798).

Baek et al. (2022) found that participants in a study are more willing to donate if an 'AI agent' is 'smiling' and looks 'like humans rather than like robots'. They found that participants 'tend to donate more in reaction to appeals from humanlike (vs. machinelike) AI agents that smile broadly rather than slightly or not at all'.

Bower and Landreth (2001) found that a 'negative affect stemming from comparisons with (...) highly attractive models may have a negative impact on advertising effectiveness'. Yang et al. (2023) examined the effects of 'ad retouching and disclaimers on advertising effectiveness' in online experiments with Chinese female participants. They found that a 'preference for flawless skin still drives purchase intentions toward a skincare brand'. A 'retouch-free disclaimer' in an ad depicting a model with realistic skin could increase the 'honesty' of an advertisement and 'led to higher purchase intentions'.

Choi et al. (2023) investigated the moderating role of social distance by comparing celebrities to Social Media Influencers (SMI). Both were differed by attractiveness and expertise. They found that 'attractiveness is more effective when the perceived social distance from the endorser is close, whereas expertise is better when the social distance is far'.

Grigsby et al. (2022) examined 'narrative transportation' achieved through 'a single image'. They found that 'single-image, picture-based ads' can lead to 'more positive attitudes toward that ad'.

2. Emotions

Overall, the study of the effect of a photo on behavior seems to touch on the area of 'emotions'. This field is not so much discussed in economic research. There are numerous articles on this area in psychology that cannot be discussed in their entirety here.

Västfjäll and Slovic concluded (Västfjäll & Slovic, 2013, p. 266) that 'emotion is now seen as a prerequisite for good decision making'. When creating a holistic model of preference formation, 'emotions' need to be discussed. What are emotions? LeDoux and Brown (2017) defined that 'subcortical circuits ... provide nonconscious inputs that coalesce with other kinds of neural signals in the cognitive assembly of conscious emotional experiences'.

According to Scherer (2005), emotions consist of five components (1) cognitive appraisal, (2) bodily symptoms, (3) action tendencies, (4) expression, and (5) feelings. Aristotle (2004, p. 142 ff) named ten specific emotions to be studied. Positive emotions are 'calm, friendship, favor, and pity'. Negative emotions are 'anger, fear, shame, indignation, envy, and jealousy'. Ekman (1992) defined six 'basic emotions': anger, disgust, fear, happiness, sadness, and surprise. Cordaro et al. (2018) found several other emotions such as 'awe', 'triumph', 'relief', 'sympathy' etc.

Hausman (2012) concluded that 'emotions' may be a source of preferences.

3. Economic thoughts on preferences

Economists take a different approach to analyzing decision making. Kirchgässner (2008, p. 12) described a person's decision-making process as follows: '[T]he individual assesses the various alternatives at his

disposal, he weighs up the pros and cons, the costs and benefits of the alternatives against each other and finally chooses that (those) alternative(s) which come(s) closest to his preferences or which promise(s) to bring about the maximum net benefit'. In many cases, the decisive factor for the decision of an individual is a so-called 'expected utility'. Expected utility is determined by the outcome of an alternative and the probability that the outcome will occur. Savage (1972) defined an outcome as a 'state'. How can a photo influence a decision in this world? A photo can either have an impact on the 'state' or an impact on the probability of this 'state'. But how would this work, once both variables, the outcome, and the probability of the state, are primarily numerical, but a photo is primarily non-numerical?

In his best-selling book, Kahneman (2011) outlined 'two modes of thinking' that he called 'systems'. 'System 1' is 'experimental' (fast, emotional) thinking. 'System 2' is 'analytic' (slow) thinking. Kahneman (2011, p. 21) described System 2 as the 'conscious, reasoning self that has beliefs, makes choices, and decides what to think about and what to do'. System 1 is, on the other hand, described with 'effortlessly originating impressions and feelings that are the main sources of the explicit beliefs and deliberate choices of System 2'. 'Humans' rely on System 1 which triggers 'several errors of intuitive judgement and choice' (Kahneman 2011, p. 416).

A photo might have a place in Kahneman's System 1, which represents emotions. In this respect, Kahneman's approach helps to move beyond the focus on 'gambling hall' (Machina, 2003) events. It allows for the measurement of 'emotional influence', which Hausman (2012, p. 118) identified as one of four sources of preferences. This opens the possibility of analyzing non-numerical information as a source of decision-making.

Galperti (2019) called on economists to develop a general theory of 'how decision makers change worldviews'. He called for a 'theory of persuasive rhetoric' that goes 'beyond Bayesianism'. Such an approach would require processing non-numerical information. Kahneman (2011, p. 202) outlined the process of an experimenter changing people's minds: 'Choosing a topic on which minds are not completely made up, (...) the experimenter measures people's attitudes. Next, the participants see or hear a persuasive pro or con message. Then the experimenter measures people's attitudes again; they usually are closer to the [more] persuasive message...'

4. The experiments – the impact of non-numerical signals on decisions

Based on my book "Media, Propaganda and Public Choice" (1997), I developed an experimental design (Jenöffy-Lochau, 2023) to measure the influence of information on decisions. I conducted experiments at a university. The participants, primarily business students, were exposed to four information signals. Additionally, the participants were exposed to non-numerical information, two photos.

When designing the experiment, the central approach was to exclude factors other than the photo in influencing the decision. The impact of the photo should be made as measurable as possible. To do this, I worked with fictional films. These films do not exist. Therefore, the test subjects' first contact with the content of these films is the experiment. The second contact is the subsequently delivered photo. This test arrangement ensures that the queried decision can be traced back exactly to the information immediately received. There are no further signals. In this respect, it is possible to measure the influence of the photo on the decision in pure form without any interference.

I initially provided 181 (2019), and then 184 students (2020) with different information. All students received four signals to describe fictitious movies. These movies were called 'The Grandfather I never wanted', 'The Mystery of the Oaktree', 'Zeppelin Jr', and 'Aberystwyth'. The content of the movies was described in one sentence.

The Grandfather I Never Wanted (π_1): *A cool Jewish guy from California finds out that his grandfather was fathered during WW2 with Hitler's genetic material. A shock with dramatic consequences.*

The Mystery of the Oaktree (π_2): *A ruthless investment banker from New York City comes to Germany to „restructure“ a company that means create bonuses for himself. An old jinx destroys him.*

Zeppelin jr (π_3): *A crooked executor of a will has a wealthy grandfather incapacitated to steal his fortune from his underage grandson.*

Aberystwyth (π_4): *A fraudulent director of a famous fashion brand tries to destroy his company from inside. A private investigator narrowly escapes him.*

A photo showed Elyas M'Barek, one of Germany's biggest movie stars. His Wikipedia entry is available in 25 languages. The photo was attached to the movie 'The Mystery of the Oaktree'. Another photo showed Pamela Reif, an Instagram personality (9.1 million followers on Instagram as of March 2025) known more for fitness videos than as an actress. The photo was attached to the movie 'Aberystwyth'.

The participants received a Questionnaire 3 with the above description of the movies. They were asked to rank the movies from '1' (most interesting) to '4' least interesting.

A Questionnaire 3a of the experiments showed photos of potential actors, M'Barek and Reif, appearing in the film. Both photos were similar in size.

In Experiment 1 (2019), the participants were divided in two random groups. Group 1 received the description of the four movies without a photo. Group 2 received the two photos. The students rated the movies from '1' (most interesting) to '4' least interesting. In Experiment 2 (2020), all participants received the additional photos a few minutes after they had rated the movies, and they were asked to rate the movies again.

4.1. Results of experiment 1 (2019)

Table 1 shows the rankings of the movies by the participants. 'The Grandfather I Never Wanted' received 37 top spots in the group without photos (Group 1) and, also 37 top spots in the group that received the additional photos (Group 2). 'Aberystwyth' had 24 top placements with no photos and only 16 top placements with the photo handed.

I added points to the ranking. A top placement received five points. A runner-up received three points. A third-place finisher received one point and a last-place finisher received no point. Table 1 shows how many points the movies received and what percentage this represents of the total number of points.

Table 1 shows an impact of the photos on the interest of the students in the respective movie. 'Aberystwyth' dropped from 24 top positions to only 16 top positions and in the end only achieved 23.7% of the total points instead of 29.2% previously.

Apparently, the photo of Pamela Reif has reduced the interest in 'Aberystwyth'. Elyas M'Barek's photo, on the other hand, increased the interest in 'The Mystery of the Oaktree' from 23.1% to 24.9% of the points. Overall, an impact of the 'emotional signal' photo was measurable. The photo changed the ranking of the movies from A>D>B>C to A>B>D>C.

4.2. Results of experiment 2 (2020)

In Experiment 2 (2020), all participants received the additional photos a few minutes after they had rated the movies, and they were asked to rate the movies again.

Experiment 2.A shows the ranking of the participants with no photos handed over. Experiment 2.B shows the results with the two photos (Table 2).

I measured an impact of the photo on the interest of the participants in the movie. 'Aberystwyth' fell from 49 top places to only 32 top places and in the end only achieved 21.1% of the total points instead of 28.5% before. Here too, the photo of Pamela Reif apparently diminished the interest in 'Aberystwyth'. On the other hand, the photo of Elyas M'Barek increased the interest in 'The Mystery of the Oaktree' from

Table 1. Ranking of the movies with, and without the additional signals in experiment 1.

Movie	Group	No.1	No.2	No.3	No.4	Points	%
The Grandfather I Never Wanted	1	37	16	18	19	251	30.9
	2	37	18	16	10	255	34.7
The Mystery of the Oaktree	1	17	27	22	25	188	23.1
	2	19	20	28	15	183	24.9
Zeppelin jr	1	11	16	33	29	136	16.7
	2	10	17	21	35	122	16.6
Aberystwyth	1	24	34	16	16	238	29.2
	2	16	26	16	23	174	23.7

Table 2. The ranking of the movies in experiment 2.

Movie	Experiment	No.1	No.2	No.3	No.4	Points	%
The Grandfather I Never Wanted	2.A	76	32	23	50	499	30.3
	2.B	78	31	26	34	509	33.9
The Mystery of the Oaktree	2.A	40	43	59	40	388	23.5
	2.B	37	56	46	28	399	26.6
Zeppelin jr	2.A	19	47	56	61	292	17.7
	2.B	20	43	48	53	277	18.4
Aberystwyth	2.A	49	60	45	29	470	28.5
	2.B	32	37	47	51	218	21.1

23.5% to 26.6% of the points. Overall, an impact of the non-numerical signal was measurable. The photo changed the ranking of the movies from $A > D > B > C$ to $A > B > D > C$ as it did in Experiment 1.

In Experiment 1, the chi-square statistic for a 2×2 contingency table was 6.79. The p value was .079115. Both variables (film/photo) were significant at $p < .1$. In Experiment 2, the chi-square statistic for a 2×2 contingency table was 73.60. The p value was .000001. Both variables were also significant dependent at $p < .01$. The results were maintained with the Yates correction.

4.3. A Bayesian analysis of the results

How would Bayesian economics explain the empirically measured results? A posterior probability of the movie 'The Mystery of the Oaktree' must have increased. The question is what preferences are involved? Do preferences for the movie 'The Mystery of the Oaktree' exist? Or, do the participants have preferences for 'a movie worth to watch?' Are there preferences for 'being entertained?' For example, Bayes would assume, constant preferences for 'a movie worth to watch'. The movie 'The Mystery of the Oaktree' is worth to watch at the prior probability is $p(\text{OAK})$. The posterior probability $p(\text{OAK} | M'B)$ is the probability of 'The Mystery of the Oaktree' worth to watch given the information Elyas M'Barek appears in that movie. It is

$$p(\text{OAK} | M'B) = \frac{p(M'B | \text{OAK})}{p(M'B)} * p(\text{OAK}) \quad (1)$$

I can measure the prior probability and the posterior probability, and thus evaluate the impact of the photo on the probabilities. If the probability of Elyas M'Barek appearing in 'The Mystery of The Oaktree' ($p(M'B)$) is given exogenously, I can determine $p(M'B | \text{OAK})$ from the data. Bayes observes a changed posterior probability. Bayes' focus, however, is not to explain why the probability has changed. The calculation gets more complicated if there are multiple observations. If you want to market 'The Mystery of the Oaktree', working with the factor $\frac{p(M'B | \text{OAK})}{p(M'B)}$ does not make it easy to depict the exact effects of your communication.

That is why I decided to show the influence of the photo on the decision more clearly using a different mathematical approach.

4.4. Extracting the impact of the photos

The modified mathematical approach aims to assign an information value to a signal. This information value can be used to predict what influence the signal will have on the decision. Visually, this looks much simpler than the mathematically much more complex Bayes method but this should not be a disadvantage. It is a way to make the influence of a non-numerical signal on a decision mathematically operational.

Thus, I will use the measured numbers to go beyond Bayesianism. I view the decision process as a seesaw. Applying the seesaw works in three steps. First, I define the preferences and information signals involved. Second, I transform the data into a $[0,1]$ interval. Yaari (1987) used such an interval for his Dual Theory. Third, I determine information values of the received information.

In both experiments, there are two relevant alternatives and two types of information signals. I define: Alternative 0 is 'The Mystery of the Oaktree'. Alternative 1 is 'Aberystwyth'. The preference space stretches from 0 to 1. At 0, there are strong preferences for 'The Mystery of the Oaktree'. At 1, there are strong preferences for 'Aberystwyth'. Figure 1 shows the seesaw of this decision-making process.

Information signal type π_0 is 'The Mystery of the Oaktree is more interesting'. The information value (ω_0) of information signals supporting this alternative is smaller than .5 ($\omega_0 < .5$). Signal type π_1 is 'Aberystwyth is more interesting'. The information value (ω_1) of information signals supporting this alternative is greater than .5 ($\omega_1 > .5$).

The fulcrum the seesaw is at .5. There, the participants are indifferent between both alternatives. Information on the left side is a 'Pro'. The heavier the 'Pro', the more the seesaw leans to the left. The information on the right side is a 'Con'. The heavier the 'Con', the more the seesaw leans to the right. If the information received would push the seesaw strongly towards 'Aberystwyth', this would mean the information value of that information is 1. Conversely, information that strongly supports 'The Mystery of the Oaktree' would have an information value of 0. The most extreme outcome of the experiment would be that 'Aberystwyth' ranked first among all participants and 'The Mystery of the Oaktree' ranked fourth among all participants took place. Then, 'Aberystwyth' had 100% of the points. P_N would be 1. All ω would be 1. The strictest outcome favoring 'The Mystery of the Oaktree' would be for all participants to have it ranked first. Then, P_N would be 0. All ω would be 0. What did I measure?

Experiment 1: I analyzed the movies 'The Mystery of the Oaktree' and 'Aberystwyth'. Without the photos, both movies received 426 points. Among that, 'Aberystwyth' received 55.9% of this share. 'The Mystery of the Oaktree' received 44.1% of that share. With the photos, both movies received a total of 357 points. 'Aberystwyth' received 48.7% of the points. 'The Mystery of the Oaktree' received 51.3% of the points. I measured $\bar{P}_{Text} = .559$ without photos, and $\bar{P}_{photos} = .487$ with photos. Table 3 above shows the results of Experiment 1.

Experiment 2: Without the photos, both movies received a total of 858 points. 'Aberystwyth' received 54.8% of this share. 'The Mystery of the Oaktree' received 45.2% of that share. With the photos, both movies received a total of 617 points. 'Aberystwyth' received 35.3% of the points. 'The Mystery of the Oaktree' received 64.7% of the points. I measured $\bar{P}_{Text} = .548$ without photos, and $\bar{P}_{photos} = .353$ with the photos. Table 4 above shows the results of Experiment 2.

Summing up, the photos shifted the decision of the participants. Non-numerical information triggered a change in the decision.

4.5. The information value of the photos

To measure the impact of the photos and use it for future predictions, I extract an information value of the photos. The information value measures the average impact of the photos on the participants' decision.

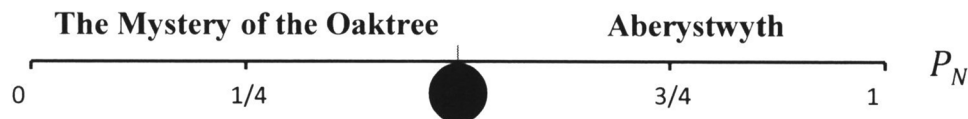


Figure 1. Preferences for the two movies.

Table 3. The shift of the decision by the photos. Experiment 1.

Movie	Group	No.1	No.2	No.3	No.4	Points	%
The Mystery of the Oaktree	1	17	27	22	25	188	44.1
	2	19	20	28	15	183	51.3
Aberystwyth	1	24	34	16	16	238	55.9
	2	16	26	16	23	174	48.7
Total	1					426	
	2					357	

Table 4. The shift of the decision by the photos. Experiment 2.

Movie	Group	No. 1	No. 2	No. 3	No. 4	Points	%
The Mystery of the Oaktree	2.A	40	43	59	40	388	45.2
	2.B	37	56	46	28	399	64.7
Aberystwyth	2.A	49	60	45	29	470	54.8
	2.B	32	37	47	51	218	35.3
Total	2.A					858	
	2.B					617	

Table 5. Preferences measured.

	Signals	P
Experiment 1		
No photo	2	.559
Photo	4	.487
Experiment 2		
No photo	2	.548
Photo	4	.353

To do this, I determine an average information value of the texts and the photos, after measuring both signals together. The average information value of the texts is $\bar{\omega}_{\text{Text}}$. The average information value of the photos is $\bar{\omega}_{\text{photos}}$. I work with the following assumptions:

- The participants had no information about the movies prior to the experiments. I ensure this with working with fictitious movies.
- The information signals ω_1 and ω_3 regarding 'The Grandfather I Never Wanted' and 'Zeppelin jr' had no influence on the preferences for 'The Mystery of the Oaktree', and 'Aberystwyth'. Therefore, this information is neglected to reduce complexity.

Table 5 shows the preferences measured in the Yaari interval.

I assume:

- Both photos have the same impact.

I determine the average information values of the texts $\bar{\omega}_{\text{texts}}$ as the initial preferences $\bar{P}_{\text{Text}}$. The average information values of the photos $\bar{\omega}_{\text{photos}}$ depend on the change between the initial preferences $\bar{P}_{\text{Text}}$ and the posterior preferences $\bar{P}_{\text{photos}}$. It is

$$\bar{\omega}_{\text{text}} = \bar{P}_{\text{text}} \quad (2)$$

$$\bar{\omega}_{\text{photos}} = 2 * \bar{P}_{\text{photos}} - \bar{P}_{\text{text}} \quad (3)$$

Table 6 shows the aggregated information values measured in the experiments. In Experiment 1, the average information value of the two photos was .415 which shifted the preferences toward 'The Mystery of the Oaktree'. In Experiment 2 the average information value of the photos was .158. The results of both experiments were similar. The two photos changed the behavior of the participants.

The Seesaw Model can depict the impact of the photos on the preferences of the individuals. The impact of the photos was stronger in Experiment 2.

5. Implications

In Experiment 2, I measured an average information value of the photos of .158. Its impact was due to a waning interest in Aberystwyth rather than an increasing interest in The Mystery of the Oaktree. Aberystwyth lost more than half of its points after participants were exposed to the photos. This is

Table 6. Average information values measured in the experiments.

	Signals	P	$\bar{\omega}_{\text{texts}}$	$\bar{\omega}_{\text{photos}}$
Experiment 1				
No Photo	2	.559	.559	–
Photo	4	.487	.559	.415
Experiment 2				
No Photo	2	.548	.548	–
Photo	4	.353	.548	.158

surprising because with millions of followers on Instagram, Pamela Reif would generally be considered an attractive woman. By determining an information value, it is possible to predict the influence of a photo on preferences.

Before one can shape or change the preferences of individuals, one must first analyze the individual's existing preferences.

- How stable are the individual's existing preferences?
- How many signals are the individual's existing preferences based on?

A producer is unlikely to face stable preferences if the movie is a new production. There may be stable preferences among individuals as to whether they want to go to the cinema in general. Relatively stable preferences could also exist for sequels or remakes. In 2019, three of the four highest-grossing movies in the US were sequels ('Avengers: Endgame', 'Toy Story 4', 'Frozen II').

To influence individual preferences, several important variables must be considered

- Will the individuals perceive my signals?
- How many signals can I transmit?
- Will the individuals receive conflicting signals from another party?

In the movie business, there will be a second type of signals that appear after people have watched a movie. People will talk about how they liked the movie. These signals may be stronger than the signals sent by a producer because they come from friends or acquaintances of the people they trust. Therefore, these signals have higher credibility than advertising signals. Signals may go 'viral'. The photo of Elyas M'Barek as the main actor also has a high level of credibility. People trust M'Barek because he is so successful that they either just want to watch him or trust him that he would only sign for an exciting script.

Some implications for the film business were first described here. However, the implications go far beyond that. In every industry, in almost every sales process, a photo can play an important role. Photos can change decisions everywhere. The mathematical method shown can be used to determine the information value of a photo in a test group. Using this information value of the photo, the influence of the photo can be projected on larger groups.

The photo's ability to influence also depends on the strength of previous decisions. If previous decisions are based on many experiences that have already been made, even photos with high information value have less chance of influencing the decision. However, if the experiences are unstable, i.e. correspond to a previous value of around 0.5, a photo has a good chance of having an impact.

6. Critics

The study produced stable results. The study sample was approximately twice as large as that used by Kahneman and Tversky (1979) when they developed prospect theory. I tested other point distributions (4/3/2/1), (5/3/2/0). They gave similar results. I chose the distribution (5/3/1/0) because it seemed to make the most sense for the topic of film selection. The information values measured depended on the participants. Different participants could lead to different information values. The average information

value is always the average information value of the respective target group. It is simply an approximation of the information value of the entire population.

7. Conclusion

I exposed students at the Humboldt University of Berlin to various information signals. They had to rate how interesting the content of four fictional films seemed to them. I conducted two experiments. In Experiment 1, one group received additional information about two actors who supposedly starred in one of the respective films. In Experiment 2, all participants received the photos after they had already answered the questionnaire to answer it a second time. I found that the photos made a significant difference in the behavior of the participants. One of the films fell in the ranking, another rose. I performed a non-Bayesian analysis of the results. I have determined an aggregated information value of the photos. With this variable it is possible to predict the impact of future communication on consumer behavior. This approach is a first step towards an economic theory of rhetoric. This approach significantly reduces the complexity of the model and makes it easier to use the research results.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Consent to participate

At the beginning of the survey, Prof. Dr. Maiterth asked the students whether they were interested in taking part in a scientific study. They were interested and the questionnaires were filled out. The study does not publish any data protected under the GDPR on any of the participants.

Ethical approval

The ethics committee of the Büro am Carlsplatz chaired by Dr. Alan Bullock has reviewed the data collection and found that there were no objections to the data collection in the case of the paper 'The Impact of a Photo on Decisions'. All participants in the study were asked verbally whether they wanted to take part in the study. Participation was anonymous. If a participant had not wanted to take part, this could have been easily done by not submitting the questionnaire. To the best of our knowledge, there is no national law in Germany that requires a written consent.

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About the author

Dr. Marek Jenöffy-Lochau holds a doctorate in economics (Dr. rer. pol.) and a degree in industrial engineering. He is managing director of Büro am Carlsplatz and a member of the board of F&P Executive Solutions AG. His research interests focus on the influence of information on decision-making.

Data availability statement

The data can be accessed at: Mendeley Data, V1, doi: 10.17632/bvm7k9mjp8.1. The data that support the findings of this study are available from the corresponding author, Dr. Marek Jenöffy, upon reasonable request.

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